ModelIt!

Teacher's Guide

# Why Do Some Seeds Travel Far?

The Amazing Ways Seeds Move to New Places

## Quick Reference

| | |
|---|---|
| **Grade Level** | Grade 2 |
| **NGSS Standard** | 2-LS2-2 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |

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# Table of Contents

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1 NGSS Standards Unpacking & CAST Alignment ..... 3

2 Scoring Rubric ..... 4

3 Background Content Knowledge ..... 5

4 The LEVER Framework ..... 6

5 Slide-by-Slide Facilitation Guide ..... 7

6 Answer Key & Expected Responses ..... 9

7 Collaborative Learning Protocols ..... 10

8 Differentiation & Extensions ..... 11

9 Common Misconceptions ..... 12

# 1 NGSS Standards & CAST Alignment

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**Performance Expectation: 2-LS2-2**

Develop a simple model that mimics the function of an animal in dispersing seeds or pollinating plants.

## Three-Dimensional Learning

| | |
|---|---|
| **Science Practice** | **Developing and Using Models** — Students develop a model showing how seed shape affects how far seeds travel, mimicking real seed dispersal. |
| **Disciplinary Core Idea** | **LS2.A Interdependent Relationships in Ecosystems** — Plants depend on animals for pollination or to move their seeds around. Seeds have features that help them move to new locations. |
| **Crosscutting Concept** | **Structure and Function** — Students connect the shape (structure) of a seed to how it moves (function) — fluffy shapes float, hooks stick, and wings spin. |

## CAST Testing Alignment

**MULTIPLE CHOICE:**

A dandelion seed has a fluffy top and floats on the wind. A coconut is heavy and falls to the ground. Which seed will travel farther by wind and why?

**CONSTRUCTED RESPONSE:**

Explain how the shape of a seed helps it travel to a new place. Give an example of a seed that uses wind to travel and describe what makes its shape special.

# 2 Scoring Rubric

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### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

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**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Plants are rooted in one spot, but their seeds can travel incredible distances. Understanding seed dispersal helps us learn how forests grow, how plants spread to new areas, and why seed shapes are so amazingly diverse.

## Why Seeds Need to Travel

If all seeds fell right under the parent plant, they would be too crowded. They would compete for sunlight, water, and nutrients. Seed dispersal spreads seeds to new locations where they have a better chance of growing into healthy plants. Some seeds travel just a few feet, while others cross entire oceans!

## Wind Dispersal

Many seeds have special shapes that help them fly. Dandelion seeds have fluffy parachutes that catch the wind. Maple seeds have papery wings that spin like helicopters. These shapes increase the seed's surface area so air pushes on them and keeps them floating longer, carrying them farther from the parent plant.

## Animal and Water Dispersal

Some seeds have hooks or sticky coatings that grab onto animal fur and get carried to new places. Some seeds are inside tasty fruits — animals eat the fruit and the seeds come out later in a new location. Other seeds, like coconuts, can float in water and travel to distant shores.

# 4 The LEVER Framework

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### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L** **Locate the System**
Students identify seed shape and wind interaction as external components and travel distance as an internal component that changes based on how the seed interacts with wind.

**E** **Establish Relationships**
Students determine that fluffy or winged seed shapes increase travel distance (positive), while heavy round shapes decrease travel distance.

**V** **Visualize & Model**
Students build a model showing how seed shape affects how far a seed travels in the wind.

**E** **Evaluate Outcomes**
Students run fluffy seed, heavy seed, and winged seed scenarios to compare travel distances.

**R** **Revise & Extend**
Students add animal carriers or water flow to explore other ways seeds disperse beyond wind.

## Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

| | |
|---|---|
| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes. |
| **As a “what-if” tool** | When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

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## Slide 1: Cover

2 min

**Say:** *"Have you ever blown on a dandelion and watched the seeds fly away? Where do they go?"*

**Watch For**

If possible, bring in a real dandelion puff or show a video of seeds floating in the wind.

## Slide 2: What Will We Learn Today?

3 min

**Say:** *"Today we are going to find out how seeds travel to new places even though plants cannot walk!"*

**Watch For**

Read objectives together. Introduce seed dispersal, adaptation, and dispersal agent with real seed examples.

## Slide 3: The Big Question

3 min

**Say:** *"Plants are stuck in the ground — they cannot move! So how do their seeds get to new places?"*

**Watch For**

Show students 3 real seeds (dandelion puff, maple wing, acorn). Ask: How do you think each one travels?

## Slide 4: Let's Build a Model!

3 min

**Say:** *"Let us build a model to figure out which seeds can travel the farthest and why."*

**Watch For**

Preview LEVER steps. Explain that we will test different seed shapes to see how they move.

## Slide 5: Activity 1: Sort the Parts

7 min

**Say:** *"What are the parts of our seed travel system? Which parts can we control?"*

**Watch For**

Guide sorting: Seed Shape and Wind Interaction are external (we choose them), Travel Distance is internal (it depends on the others).

## Slide 6: Activity 2: Connect with Arrows

7 min

**Say:** *"If a seed is fluffy, will it travel far or not far in the wind? Let us draw our arrows."*

**Watch For**

Help students connect Seed Shape and Wind Interaction to Travel Distance with positive arrows for fluffy/winged shapes.

## Slide 7: Activity 3: Run the Simulation!

8 min

**Say:** *"Time to test! We will drop a fluffy seed, a heavy acorn, and a winged seed and see which goes farthest."*

**Watch For**

Students predict first. Run simulations (or drop real model seeds). Measure and compare travel distances.

## Slide 8: What Did We Discover?

7 min

**Say:** *"Which seed won the travel contest? Why did that shape help it go the farthest?"*

**Watch For**

Discuss how seed shape is an adaptation. Connect to why plants need seeds to travel — room to grow!

# 6 Answer Key & Expected Responses

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## Component Sorting

- **EXTERNAL:** Seed Shape, Wind Interaction
- **INTERNAL:** Travel Distance

## Relationships

- **Seed Shape (fluffy/winged) to Travel Distance:** POSITIVE (+) — Seeds with fluffy or winged shapes catch more air and float longer, so they travel farther from the parent plant. The more surface area the seed has, the more the wind can push it along.
- **Wind Interaction to Travel Distance:** POSITIVE (+) — Stronger wind or better wind-catching seed shapes mean the seed travels farther. More wind energy helps carry the seed a longer distance before it lands.

## Key Discoveries

- Seeds with fluffy or winged shapes travel farther in the wind than heavy round seeds
- The shape of a seed is an adaptation that helps it get to a new place to grow
- Different seeds use different methods — wind, water, animals, or even popping open
- Seeds need to travel away from the parent plant so they have enough room, sunlight, and water to grow

# 7 Collaborative Learning Protocols

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### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
|---|---|
| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

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### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

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### Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "Seeds are planted by people or they do not grow" | Most seeds in nature are NOT planted by people! Seeds have their own ways of getting to new places — wind, water, animals, and even exploding pods. Forests and meadows grew from seeds that dispersed naturally. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "All seeds travel the same way" | Different seeds have very different shapes because they travel in different ways. Dandelion seeds fly on the wind, burr seeds hitch rides on animals, coconut seeds float in water, and some seeds even pop out of exploding pods! |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Seeds are not alive" | Seeds ARE alive — they have a tiny baby plant inside waiting to grow! Seeds look like they are not doing anything, but inside they are just waiting for the right conditions: water, warmth, and soil. Then they wake up and start growing. |